Assignment-1

Theory Part

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1. When we train a ridge regression without regularization, the Loss function is

$$\mathcal{L}(w) = \frac{1}{n}\sum_{\mu=1}^{n} (y_\mu - X_\mu \cdot w)^2$$

the minimization of this function has the form $X^T y = X^T X \hat{w}$. In the case $n < d$ the Covariance matrix $X^T X$ is not invertible. Indeed, for $n < d$ we have $\mathrm{rank}(X) < d$, which implies that the least square minimization problem has infinitely many solutions. In fact, in the case $n < d$ we have more parameters than the number of samples, i.e. the number of equations.

2. In a ridge regression with zero regularization, given the data $X \in \mathbb{R}^{n\times d}$ and the labels $y \in \mathbb{R}^n$, we have the linear system $y_\mu = \sum_{i=1}^{d} X_{\mu i}\, w_i \quad, \forall \mu = 1, \ldots, n$. Depending on (n, d) we have different kind of fitting. If $n > d$, then we will have more independent equations than parameters, so the system will not have a solution. This correspond to a fit which does not fit all the data. On the other hand, when we have $n < d$, the system has infinitely many solutions. In this case the solutions of the sistem fit the data points, but at the same time the model fit also the noise in the data. In the special case $n = d$, the system has a unique solution, which fits perfectly the training points.

3. The training error increases with the increasing of the regularization strenght λ. It has a maximum limit for large λ (i.e. $\lambda \to \infty$) corresponding to the norm of the labels vector y_{train}. Indeed:

$$E_{\text{train}} = \frac{1}{n_{\text{train}}} \sum_{\mu \in \text{train set}}^{n_{\text{train}}} (y_\mu - X_\mu \cdot w)^2$$

But necessarily the parameters $w \to 0$ for $\lambda \to \infty$, otherwhise, for any finite value of w, the Loss function $\mathcal{L}(w) = \frac{1}{n}\sum_{\mu=1}^{n} (y_\mu - X_\mu \cdot w)^2 + \lambda \sum_{i=1}^{d} w_i^2$ would diverge to infinity.This implies that:

$$E_{\text{train}} \to \frac{1}{n_{\text{train}}} \|y_{\text{train}}\|^2$$